

PHARMACOVIGILANCE • TASK 4

Safety Signal Review Capstone

Professional signal-assessment briefing based on a small synthetic aggregate adverse drug reaction dataset

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Project type	Pharmacovigilance safety signal review
Dataset	8 synthetic ADR case reports
Primary signal assessed	Warfarin — uncontrolled hemorrhage
Assessment date	September 2026

Important interpretation: This review identifies a potential signal for further evaluation. It does not establish causality, incidence, relative risk, or a confirmed safety concern.

Prepared as a portfolio-style pharmacovigilance exercise using the synthetic aggregate dataset supplied with the task.

1. Executive Summary

Eight synthetic adverse drug reaction (ADR) cases were reviewed descriptively. Each case involved a different suspected medicine and adverse reaction. Patients ranged from 28 to 71 years, with an equal male/female split. The dataset includes mild through life-threatening events and outcomes ranging from recovery to hospitalization and non-recovery.

Metric	Finding
Total cases	8
Age range	28–71 years
Mean / median age	52.0 / 51.5 years
Gender	4 female; 4 male
Reported onset	1–30 days (mean 11.2; median 8.5)
Severity	3 mild; 2 moderate; 2 severe; 1 life-threatening
Outcomes	3 recovered; 1 recovering; 3 hospitalized; 1 not recovered
Reporter type	3 physicians; 2 pharmacists; 2 patients; 1 oncologist

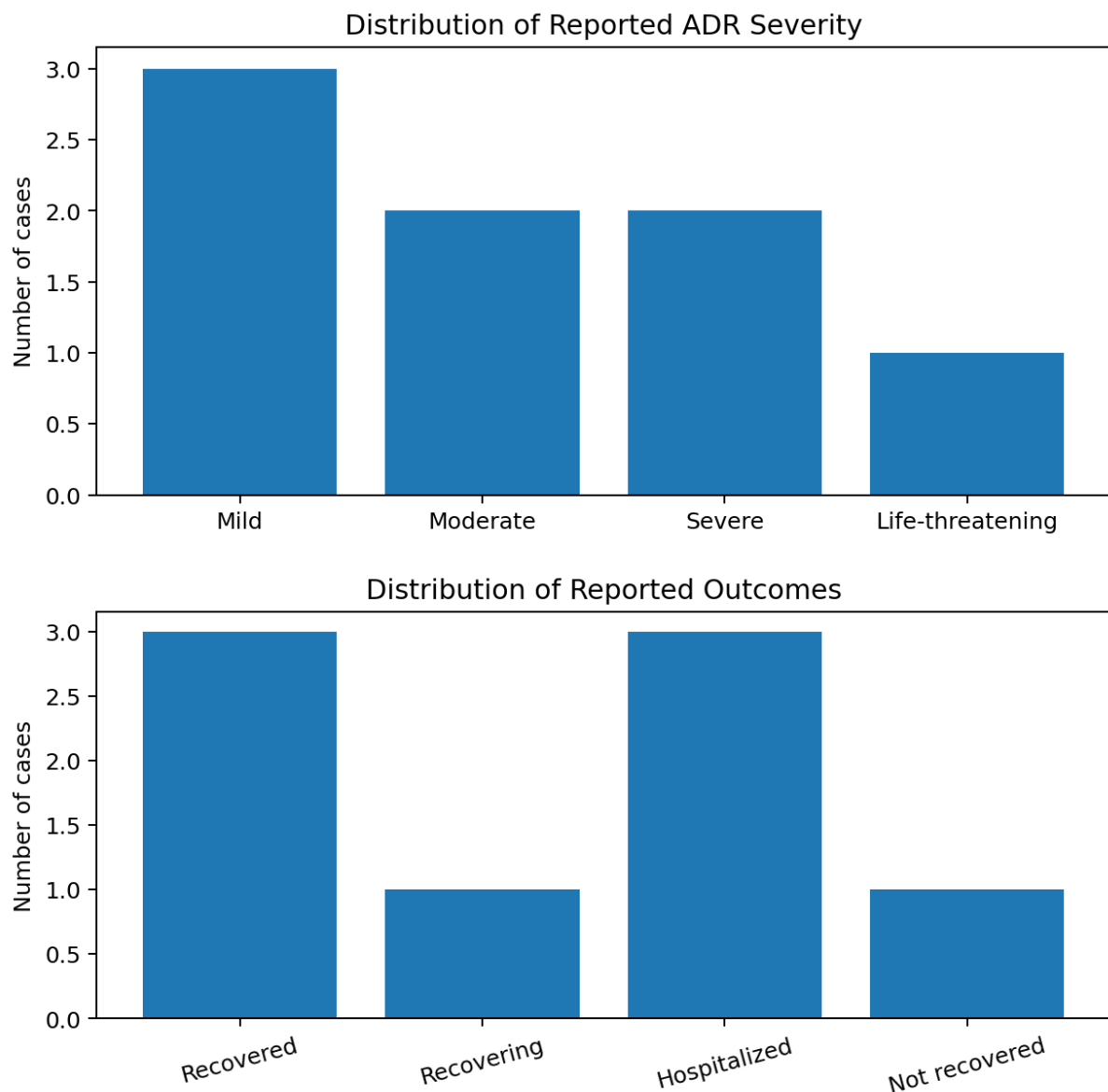
2. Case-Level Dataset Review

Report ID	Age	Sex	Drug	Adverse reaction	Onset	Severity	Outcome	Reporter
ADR-9001	45	F	Amoxicillin	Skin rash & hives	2 d	Mild	Recovered	Pharmacist
ADR-9002	62	M	Atorvastatin	Severe muscle myalgia	14 d	Moderate	Recovering	Physician
ADR-9003	71	F	Ciprofloxacin	Tendonitis	7 d	Severe	Hospitalized	Physician
ADR-9004	28	M	Ibuprofen	Gastric ulcer bleed	5 d	Severe	Hospitalized	Patient
ADR-9005	53	F	Lisinopril	Persistent dry cough	21 d	Mild	Recovered	Pharmacist
ADR-9006	39	M	Metformin	Nausea & diarrhea	1 d	Mild	Recovered	Patient
ADR-9007	68	M	Warfarin	Uncontrolled hemorrhage	10 d	Life-threatening	Hospitalized	Physician
ADR-9008	50	F	Paclitaxel	Peripheral neuropathy	30 d	Moderate	Not recovered	Oncologist

Interpretive note: The supplied dataset is synthetic and small. Therefore, counts describe this dataset only and should not be interpreted as incidence rates or comparative risks.

3. Severity, Outcome and Reporting Trends

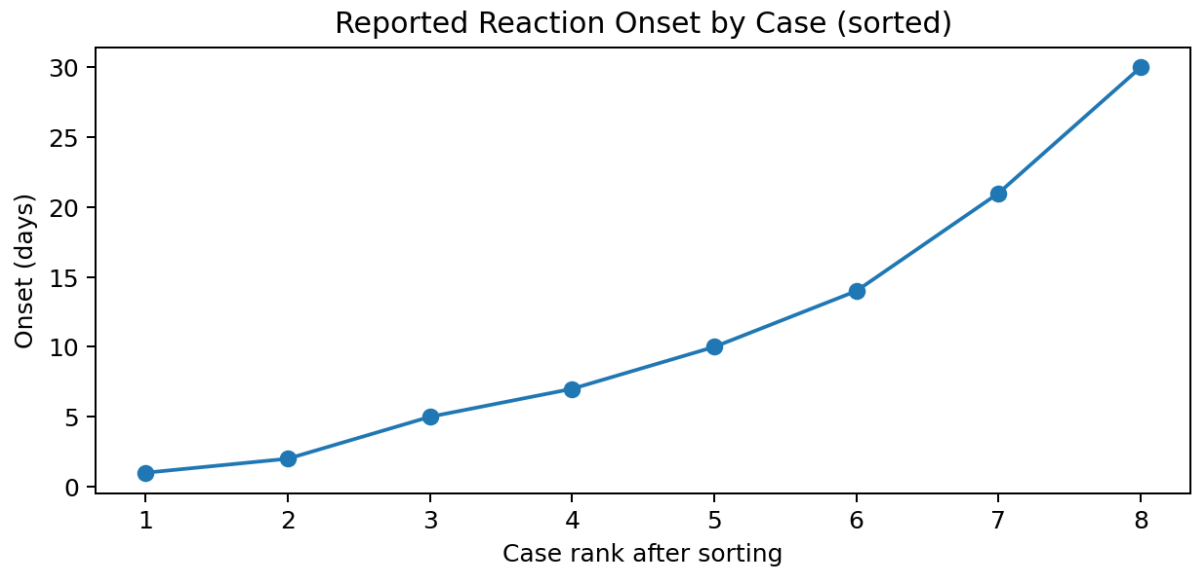
The charts below summarize the distribution of reported severity and outcomes. In pharmacovigilance, *severity* and *seriousness* should not be treated as synonyms. Regulatory seriousness is based on criteria such as death, life-threatening events, hospitalization, disability, or other medically important outcomes.



Trend observation: Three cases were labeled severe or life-threatening, and three had hospitalization as the recorded outcome. The single life-threatening case was also hospitalized. Overall, at least four cases contain a seriousness criterion based on life-threatening status or hospitalization, although this is a descriptive classification of the supplied records.

4. Reaction-Onset Pattern

Reported onset ranged from 1 to 30 days. The median onset was 8.5 days. The earliest event was metformin-associated nausea and diarrhea at 1 day, while the latest was paclitaxel-associated peripheral neuropathy at 30 days.



5. Potential Safety Signal

Drug–event combination	Warfarin — uncontrolled hemorrhage
Patient	68-year-old male
Onset	10 days after reported exposure
Severity	Life-threatening
Outcome	Hospitalized
Reporter	Physician
Signal rationale	Clinically serious event with a temporal relationship and a medically important outcome

The warfarin–hemorrhage combination is prioritized for further review because the event is reported as life-threatening and resulted in hospitalization. A single case can prompt further investigation when the event is clinically important; however, a signal is a hypothesis requiring additional evidence rather than proof that the medicine caused the event.

6. Alternative Explanations / Confounding Factors

Potential alternative explanation	Why it matters
Excess anticoagulant effect / high INR	Could indicate that anticoagulation intensity exceeded the intended therapeutic range.
Concomitant medicines	Drug interactions may increase bleeding risk or alter warfarin exposure/effect.
Patient-related factors	Age, comorbidities, prior bleeding, falls, or frailty may contribute to the event.
Dose or dose-change history	A recent dose adjustment or dosing error could affect temporal interpretation.
Underlying disease	The indication and underlying clinical condition may independently influence bleeding risk.
Incomplete case information	The aggregate record lacks laboratory, medication-history, and detailed clinical information.

7. Recommended Follow-up Evidence

Follow-up question	Purpose
What was the warfarin dose and treatment indication?	Characterize exposure and clinical context.
What was the INR at the time of hemorrhage?	Assess anticoagulation intensity.
What was the target INR range?	Compare measured vs intended anticoagulation.
Were there recent dose changes or dosing errors?	Evaluate a possible exposure-related explanation.
Which prescription, OTC and herbal medicines were used?	Identify possible interactions or additive bleeding risk.
What was the bleeding site and clinical management?	Characterize the event and its medical importance.
What happened after dose reduction/withdrawal?	Assess dechallenge information where available.
Was warfarin restarted and what happened?	Assess rechallenge information where available.
Was there previous bleeding or relevant renal/hepatic disease?	Identify patient-level risk factors.

8. Risk-Minimization Considerations

Question / action	Rationale
Confirm appropriate INR monitoring	Supports safe anticoagulation management and early identification of excessive anticoagulation.
Perform medication reconciliation	Helps identify interacting prescription, OTC, and complementary medicines.
Review dose and administration instructions	Reduces preventable dosing and administration errors.
Educate patients on bleeding warning signs	May support earlier recognition and clinical intervention.
Escalate serious cases for appropriate medical review	Life-threatening or hospitalized events warrant prompt clinical assessment and follow-up.

9. Conclusion

The review of eight synthetic ADR cases identified several clinically important events. The most notable potential signal was uncontrolled hemorrhage reported in a 68-year-old patient receiving warfarin, with a reported onset of 10 days, life-threatening severity and hospitalization. The observation warrants further assessment but does not establish causality. Important alternative explanations include excessive anticoagulation, interacting medicines, dose-related factors, underlying disease and patient-specific risk factors. Additional case-level information, laboratory results, medication history and larger safety datasets would be required to characterize the signal and determine whether any change in risk-management action is justified.

Overall assessment: Potential signal identified → further investigation recommended → no causal conclusion from this dataset alone.

10. Key Limitations

- Synthetic dataset with only eight cases.
- One reported case for each drug–ADR combination.
- No exposure denominator; therefore incidence or relative risk cannot be calculated.
- No complete concomitant medication, laboratory, medical-history or dose information.
- Aggregate data limit individual case causality assessment.
- Reporting and selection biases cannot be excluded.

11. References

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Portfolio note: The supplied ADR dataset is synthetic and was used only for this educational signal-review exercise.